BROKILD KITCHEN APPLIANCES · WINDOWS VST3 · OWNER'S MANUAL & COOKBOOK

Hairfryer

A vocal strip that turns a clean singing voice into a metal one — by wrecking the part of the voice a fuzz pedal can never reach.

HAIRFRYER

gate · de-esser · the basket · tube heat · EQ · compressor · limiter

BROKILD

free · VST3 + standalone · 64-bit

01 Why this should not work, and why it does

Put a distortion pedal on a voice and you get a distorted voice — which is not a metal voice at all. The reason is worth one paragraph, because it is the whole design of this plugin.

A sung note is a **source** passed through a **filter**. The source is the glottal pulse train — the folds slapping together; the filter is your throat and mouth, whose resonances (the *formants*) are what make an *ah* an *ah*. A pedal sits after the filter, so it mangles the formants and intermodulates them with each other. The vowel goes to mush, the words disappear, and the ear says “voice through a pedal”, never “monster”.

A real screamer does the opposite. The vocal tract stays — that is why you can hear the words in a good growl — and the **source** is what changes: the false folds above the true folds are driven into a period-doubled or chaotic vibration. The pitch you hear is still the true folds, which is why growlers can sing melodies. The monster lives in subharmonics at f/2 and f/3 and in turbulence fired in step with the glottal pulses.

So the Hairfryer takes your voice apart in real time — an order-18 LPC lattice splits it into glottal source and vocal tract — then **wrecks only the source** and puts your own vocal tract back on top. Everything it adds comes out wearing your formants. Your vowels. Your words.

> **Measured, not hoped.** The offline bench sings a synthetic vowel through the plugin and measures the claim by Goertzel: the input's formant-to-floor contrast is `35.2 dB`; the growled output keeps `35.0 dB` of it; the same voice through a fuzz pedal collapses to `19.7 dB`. That difference is the entire reason this instrument exists.

The chaos is real chaos

Vocal-fold dynamics is a textbook bifurcation system, and so is the logistic map `x ← r·x·(1−x)`. The CHAOS knob drives one iteration of that map per detected glottal period and turns the result into the period's gain, noise and timing. Walking the knob walks the real cascade:

KNOB	R	REGIME	WHAT YOU HEAR
0 – .09	< 3.0	steady	clean, or merely gripped
.09 – .55	3.0 – 3.57	period 2, then 4	an octave-down rasp; the classic low growl
.55 – .80	3.57 – 3.83	chaos with windows	fry — irregular, torn
.80 – .90	3.8319	**period 3**	the deep guttural — the knob has a plateau here so you can find it
.90 – 1	→ 4.0	full chaos	everything at once

The basket window on the panel shows the last forty periods of that gain pattern per rack, and names the regime it measures: STEADY, P2, P3, CHAOS.

02

Thirty seconds of instructions

1. Put the Hairfryer on a vocal channel. It is an insert effect, mono or stereo.

2. Pick a **PROGRAMME**. FALSE CORD is the low growl, GUTTURAL is deeper, PIG SQUEAL is the squeal, BLACKENED is the high torn scream. CLEAN PASS and WARM STRIP leave the fryer off — they are ordinary vocal-strip presets.

3. Sing loud, clean and supported. The fryer reads the pitch and the pulses of what you give it. A confident full-voice note gives it everything it needs; a whisper gives it nothing to wreck.

4. Ride **BLEND**. Even 40 % over the clean voice reads as “that singer has a torn edge”; 100 % is the full creature.

5. Then treat it like any vocal chain: HEAT for density, SEASONING to seat it, PRESSURE to hold it still.

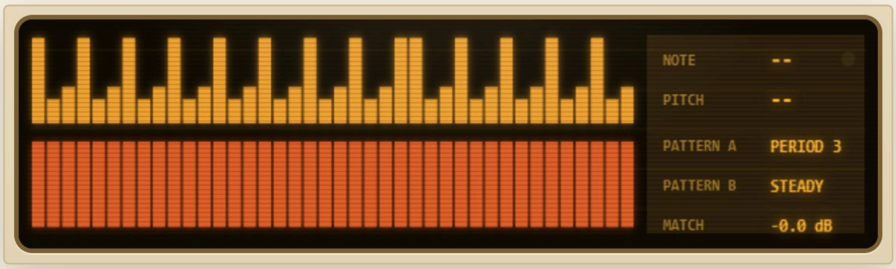
Nine programmes and the LCD. Every knob you touch is narrated there.

Level honesty. With MATCH on, the wet path is continuously gain-matched to the dry voice, so BLEND compares character, not loudness — and the growl cannot win the argument by being louder (measured: within 0.2 dB). RANDOM cooks a new basket without touching your gain structure.

Latency. The analysis, the oversamplers and the limiter's lookahead cost about 4.6 ms at 48 kHz, reported to the host, and deliberately constant — switching oversampling does not re-align your session.

03 The basket

THE PART THAT SHOULD NOT BE POSSIBLE



The basket window: forty periods of gain pattern per rack, pitch, regime, match.

Two **racks** cook in the same basket. Both are fed the same analysed source; each wrecks it its own way and wears its own vocal tract. A metal vocal on a record is nearly always more than one take stacked — RACK A is the take, RACK B is the stack. TWO THROATS runs a growl under a shriek.

CONTROL	WHAT IT ACTUALLY DOES
FRYER	the whole basket in or out (the strip stays)
BLEND	clean voice ↔ cooked voice, gain-matched
CRISP	how sharply the noise bursts hug the glottal closures — low is breath, high is gravel
MATCH	the honesty switch (see §02)

PER RACK	
LEVEL	this rack's share of the wet signal
CHAOS	position on the bifurcation cascade (§01) — the LCD shows the actual r
GRIP	how deep the period pattern carves. At zero the map runs but touches nothing.
JITTER	per-period timing wobble — roughness sidebands, age, damage
RASP	turbulence, fired at the closures, shaped by your formants, gated by the same chaos — it breaks up when the voice does
TONE	colour of that turbulence, dark to bright
FOLD	waveshapes the source itself (oversampled). The one knob here that is allowed to sound like a distortion — on the source, it reads as strain, not fuzz
SPLIT	the false folds: a resonator with a saturator in its feedback, driven by your voice. Low, it colours; high, it self-oscillates and locks to a fraction of your pitch — two voices at once
RATIO	which fraction SPLIT aims for: f/4, f/3, f/2, f, 1.5f
THROAT	resizes you. Warped-LPC formant slide, ±35 % — pitch does not move. Down is a bigger chest; up is the pig squeal

Silence stays silent. Every wrecking stage is scaled by the voice's own excitation, so the noise, the folds and the chaos all die with your note. RASP at maximum into silence measures 0.000001 RMS out.

04

The strip around it

PREP

A downward gate with hold (sing through it, it opens; stop, it closes on the spill), a fourth-order low cut, and a de-esser that judges sibilance by the *ratio* of high band to everything — a singer getting louder is not an ess and is not ducked like one. Measured: a bare ess ducked 18.8 dB while a vowel through the same settings moved 0.00 dB.

HEAT

The tube stage: an asymmetric soft clip (even harmonics — warmth), with SPARKLE driving the band above 5.5 kHz into its own gentle clip and folding it back — an exciter. Oversampled 1×/2×/4× at constant latency. The stage is RMS-matched around itself, so DRIVE changes density, not volume — and at zero it is bit-transparent, measured at 0.33 dB worst-harmonic deviation for the whole bypassed strip.

SEASONING

Four bands: low shelf, two parametric mids, high shelf. RBJ curves — the ones every engineer's ear is calibrated against. For a growl, try cutting 300–500 Hz slightly and adding 2–4 kHz: the words come forward.

PRESSURE

A soft-knee compressor with two-stage smoothing (a fast stage catches, a slow one breathes — the analogue read), auto or manual makeup; then a lookahead limiter (3 ms) with a soft ceiling behind it that is never off. Measured: 12 dB more in → 4.6 dB more out at the default ratio; the ceiling held against a 3× overdriven input.

05 Cookbook

THE NINE PROGRAMMES, AND HOW TO SING AT THEM

PROGRAMME	WHAT IT IS
CLEAN PASS	the strip alone, fryer off. A clean vocal channel.
WARM STRIP	strip pushed: tube, glue, sheen. Still a singing voice.
AIR FRY	light fry over the clean voice — a worn edge, 45 % blend.
FALSE CORD	period-2 growl, false folds locked at f/2, throat down. The classic.
GUTTURAL	the period-three plateau, folds at f/3, throat 0.35. The deep one.
PIG SQUEAL	throat shrunk hard, folds pushed above the pitch, bright.
BLACKENED	full chaos, bright pulsed noise, fold, sparkle. High torn screams.
DOOM LOW	slow, huge: f/4, deep grip, big chest, dark EQ.
TWO THROATS	both racks: growl under, shriek over.

How to sing at it

Give it pressure. The fryer scales everything by your own excitation. Support the note like you mean it; the wreckage follows your dynamics exactly.

Sing the melody straight. The pitch survives the growl (measured: f0 beats its neighbours by 40 dB). You supply the tune, the basket supplies the monster.

Vowels are yours. The formants pass through, so diction does what it does on a real growl: open vowels read huge, closed vowels read strangled. Both are idiomatic.

Doubling. For a chorus, duplicate the track, RANDOM the copy, pan both. Two different baskets on the same take is the cheap version of tracking it twice.

What it is not. There is no AI in here, no voice model, no sample of somebody else's scream. It is filtering, chaos and resynthesis, running in a few hundred operations per sample — your voice is the only voice in the room. Which also means it will not scream *for* you: it converts commitment. That is, honestly, rather metal.

Appendix

Install

Copy the whole `Hairfryer.vst3` **folder** into `C:\Program Files\Common Files\VST3\` (a sub-folder such as `\Brokild\` is fine), then rescan plugins in your DAW. It appears as an **effect** called Hairfryer, by Brokild. Or run `Hairfryer.exe` — the same thing, standalone, no install.

Presets

SAVE / LOAD write ordinary `.json` files; the PRESETS menu lists the preset folder (by default `User presets` beside the installed plugin, falling back to Documents). Presets carry all 62 parameters.

What the bench proves

Every build is run through an offline bench (`test/`) that renders a synthetic sung vowel through the engine and measures — not inspects, measures:

CLAIM	RESULT
bypassed strip transparent	`0.33 dB worst harmonic`
period-2 raises f/2	`+15.6 dB over input`
period-3 raises f/3	`+13.6 dB over input`
formant contrast kept vs fuzz	`35.0 vs 19.7 dB`
pitch beats off-pitch neighbours	`by 40 dB`
chaos knob regimes 1 / 2 / 3	`measured 1 / 2 / 3`
false folds lock at f/2	`+29 dB`
silence in, RASP max	`0.000001 RMS out`
growl vs clean loudness	`+0.2 dB`
62 params × extremes, 200 random, 4 rates	`all bounded, no growth`

Physiology, briefly

The design leans on the source-filter model of speech, on the false (ventricular) folds' role in growled phonation, and on the observation — Herzel, Titze, Berry and colleagues — that vocal-fold vibration undergoes period-doubling bifurcations into chaos exactly as simple nonlinear maps do. The logistic map here is not a metaphor; it is the same mathematics driving a different oscillator.

HAIRFRYER · Brokild Kitchen Appliances · August 2026 · Windows 64-bit VST3 + standalone · free. Built with JUCE. No warranty; the appliance has no thermostat. If you cannot find the growl, raise GRIP before raising CHAOS — the map can be running with nothing attached to it.